

Data Engineer Roadmap (2025)

A complete step-by-step roadmap to become a Data Engineer, from beginner to advanced, including tools, concepts, and projects.

Phase 1: Foundations (Programming, SQL, Linux)

Before diving into big data tools, focus on core fundamentals.

Core Skills

- **Programming Language:** Python (most preferred) or Scala/Java
 - Focus on data structures, loops, file handling, JSON, and APIs
 - Learn libraries: pandas, numpy
- **SQL:**
 - Learn joins, subqueries, window functions, CTEs, and group by
 - SQL optimization: EXPLAIN plan, indexing
 - Practice on LeetCode, StrataScratch, or Mode Analytics
- **Linux / Shell Scripting:**
 - Commands: ls, cat, grep, awk, sed, find
 - Write Bash scripts for data automation

Tools:

MySQL, PostgreSQL, VS Code, Jupyter Notebook

Phase 2: Data Modeling and ETL Concepts

Key Areas

- **Data Modeling:**
 - Understand OLTP vs OLAP
 - Star and Snowflake Schemas
 - Fact and Dimension tables
 - Normalization and Denormalization
- **ETL (Extract, Transform, Load):**
 - Understand how raw data is extracted, cleaned, and loaded into data warehouses
 - Build pipelines using Python, Pandas, or ETL tools like Airflow and ADF

Tools:

Apache Airflow, dbt, Azure Data Factory, AWS Glue, Google Dataflow

Phase 3: Cloud Platforms

Learn at least one cloud platform in depth (others conceptually).

AWS: S3, Glue, Redshift, EMR, Lambda, Athena

Azure: Data Factory, Data Lake, Synapse, Databricks, Functions

GCP: BigQuery, Dataflow, Dataproc, Composer

Goal: Build a cloud-based end-to-end data pipeline for ingestion, transformation, and analytics.

Phase 4: Big Data Ecosystem

Hadoop Fundamentals

- Understand HDFS, MapReduce, YARN
- Hive and Pig (conceptually only)

Apache Spark (Core Skill)

- RDD, DataFrame, Dataset
- Spark SQL, PySpark
- Performance tuning: Catalyst optimizer, Tungsten engine, partitioning, caching

Hands-on Projects

- Process large CSV/JSON datasets with PySpark
- Build real-time log processing using Spark Streaming

Tools:

PySpark, Databricks, AWS EMR

Phase 5: Data Warehousing and BI

Concepts:

- ETL to DWH to BI Flow
- Partitioning, bucketing, slowly changing dimensions (SCDs)

Tools:

- **Data Warehouses:** Snowflake, Redshift, BigQuery
- **BI Tools:** Power BI, Tableau, Looker

Project Idea:

Create a Sales Analytics Data Warehouse and visualize results in Power BI.

Phase 6: Streaming and Advanced Topics

Streaming Tools

- Apache Kafka (producer, consumer, topics, partitions)
- Spark Structured Streaming
- AWS Kinesis, Azure Event Hub

Workflow Management

- Build, schedule, and monitor DAGs in Airflow
- Implement retries, backfills, and SLAs

Data Governance and Quality

- Implement data validation using Great Expectations or Monte Carlo
- Understand data lineage and metadata management

Project Example:

Real-time user activity tracking using Kafka, Spark, and Power BI.

Phase 7: Real-World Projects (Portfolio Building)

1. **ETL Pipeline Project:**
Source data from APIs → Transform in Airflow → Load into Snowflake
2. **Streaming Data Pipeline:**
Kafka → Spark → Delta Lake → Power BI Dashboard
3. **End-to-End Cloud Data Pipeline:**
Azure Data Factory → Databricks → Synapse → Power BI

Tip: Upload your projects to GitHub and document them on Medium or LinkedIn.

Phase 8: System Design for Data Engineers

Learn architectural design and scalability patterns:

- Data partitioning and clustering strategies
- Lakehouse architecture (Delta Lake, Iceberg)
- Data Mesh principles
- Workflow orchestration and pipeline monitoring patterns

Resource

Pack Title	Description	Link
Data Engineer Interview Mega Pack 2025	The ultimate collection covering SQL, Spark, ADF, Cloud, and scenario-based questions. Trusted by 1000+ learners who've landed roles in top tech firms.	https://lnkd.in/gBfGiUpf
Data Engineer Interview Mix Pack 2025	100 real-time, scenario-based Q&As designed to test practical data engineering problem-solving skills.	https://lnkd.in/gu-34Ngz
600+ Real Data Engineer Interview Q&As (2025 Edition)	A massive collection of questions from top tech companies — EY, Infosys, TCS, Dell, Wipro, and more.	https://lnkd.in/gZA3D3-D
Top 100 Real Data Engineer Q&As (1–4 Years Experience)	For beginners and early-career data engineers. Covers SQL, ETL, Cloud, and real-time scenarios.	https://techinterviewtitans.com/product/top-100-real-data-engineer-interview-questions-answers-2025-edition-1-4-years-experience/
100 Real Data Engineer Q&As (4–8 Years Experience)	For mid-level data engineers. Includes advanced Spark, ADF, and cloud data pipeline scenarios.	https://techinterviewtitans.com/product/100-real-data-engineer-interview-questions-answers-2025-edition-for-4-8-years-of-experience/

Final Tech Stack (2025-Ready)

Recommended Stack:

Python + SQL + Spark + Airflow + Kafka + Snowflake + Power BI + Azure Data Factory

This stack is sufficient to land a Data Engineer or Cloud Data Engineer role in 2025.
